ORIGINAL ARTICLE

CatGrass: a feature engineering framework to forecast the seismic response of low-rise RC frames using CatBoost algorithm integrated with grasshopper optimization

Ahmet Demir[1] · Selçuk Demir[1] · Emrehan Kutlug Sahin[1]

Received: 29 January 2025 / Accepted: 20 May 2025 / Published online: 17 June 2025

Abstract

Predicting the maximum drift ratio (*MDR*) of reinforced concrete frames is a critical parameter in the design, evaluation, and rehabilitation of reinforced concrete structures. This study aims to investigate the prediction of the *MDR* of regular and irregular low-rise reinforced concrete frames using the grasshopper optimization algorithm (GOA) optimized CatBoost algorithm (i.e., CatGrass). For this purpose, a total of 2,300 ground motion records were utilized, and the dataset includes 20 different intensity measures (IMs), spectral acceleration values at the first mode period $S_a(T_1)$, as well as the results of nonlinear time history analyses for each record. Also, several feature engineering techniques (advanced ML pipeline) were applied including outlier detection using the multivariate outlier detection and replacement method (i.e., *outForest*), missing data imputation with the remove missing values function (i.e., *na.omit*), data transformation using the Box-Cox method, and feature selection via the Chi-square method. To evaluate model performance, five statistical evaluation metrics, namely R^2, RMSE, MSE, MAE, and MAPE were employed. For comparison, the prediction results of the CatGrass models were evaluated against those of the raw CatBoost model (benchmark ML pipeline) and some popular ML algorithms, which do not use any feature engineering techniques. The results demonstrated that the proposed CatGrass model achieved a higher R^2 value (exceeding 0.99) and lower MSE, RMSE, MAE, and MAPE values compared to the raw CatBoost model and the other ML models in predicting the MDR. This study confirms the effectiveness of the proposed novel CatGrass model, establishing it as a computationally efficient, time-saving, and reliable tool for seismic design and assessment of RC frames.

Keywords CatBoost · Feature engineering · Grasshopper optimization · Machine learning · Maximum drift ratio · RC frames

1 Introduction

Earthquakes are one of the most important natural disasters, causing both casualties and damage to structures. Every year, numerous major earthquakes occur worldwide, leading to serious socio-economic losses. The extent of casualties and damage to structures that result from an earthquake depend on factors such as the building stock, the seismic behavior of the structures, and the properties of the ground shaking. For these reasons, the seismic assessment of the structures is of the utmost importance in earthquake engineering. Seismic assessment of low-

https://doi.org/10.1007/s00521-025-11387-z

and mid-rise reinforced concrete (RC) frame structures is particularly crucial because they build approximately 75% of the total building stock in Türkiye and are generally built for residential and school purposes. Additionally, the ratio of low- to mid-rise such as residential and school buildings to all buildings is also high worldwide [9, 55, 69].

In recent decades, for the seismic design, evaluation, and rehabilitation of low- and mid-rise buildings, the analysis method of nonlinear time history analysis (NTA) of structures has become a common analysis method due to advances in the processing power of computers and the availability of ground motion (GM) records from the database [1, 2]. With NTAs, target response parameters of the structures such as maximum and interstory displacement, residual displacement, and rotation of the structure's members can be found more realistic (Priestley 2007; [64]). Different seismic codes such as EUROCODE-8 (EC8)[26], ASCE 07–16 [6], and TBEC [78] contain relatively similar conditions for performing NTAs. In these codes, artificial, synthetic, and real GMs downloaded from the database can be used. In addition, the mean of the structural response parameter can be used for design and/or evaluation of structures if more than seven GM records are used. It should be noted that the selection of the GM records can significantly affect the target response parameters of the buildings. Therefore, the selection of the GM records for NTA is a crucial step for the design and/or evaluation of the buildings.

It is also worth noting that the GM intensity measures (IMs) are important parameters for estimating target response parameters. Previous studies have shown a strong correlation between target response parameters and IMs [45, 18, 61,]). For example, Kostinakis et al. [45] investigated the correlation between mid-rise RC buildings and nineteen widely used IMs. In their study, four different RC buildings were considered and 64 bidirectional ground motions using for NTA. The analysis results indicated that the spectral acceleration at the fundamental period $S_a(T_1)$ of the buildings has the strongest correlation with the buildings damage. Conversely, there is a medium or poor correlation between the majority of IMs and overall building damage index. Meral [55] investigated the correlation between IMs and energy demands for low-rise RC buildings such as 4- and 7-story residential buildings without shear walls. The study used 20 different IMs and 44 strong GM records. IMs parameters related to velocity and acceleration showed stronger correlations than those related to frequency and displacement.

It should be noted that the NTAs of buildings provides a more comprehensive and accurate assessment of target response parameters. However, this analysis method is more complex and time-consuming especially for two- or three-dimensional analysis of buildings [18, 32, 36]. Therefore, researchers have needed simpler and less time-consuming computational tools for seismic assessment of the buildings. Eventually, machine learning (ML) algorithms have been applied widely in recent years for seismic assessment of the buildings. In addition, ML algorithm was used also in different areas of civil and earthquake engineering areas. For example, there is extensive research on the use of ML algorithms in civil and earthquakes engineering areas such as seismic damage and performance assessment of buildings ([19, 22, 39, 44, 46, 59]), liquefaction potential [20, 71], damage classification [52, 66], compressive strength of concrete material [63, 83], and prediction of the first periods of buildings [56, 80]. Finally, many researchers have also provided state-of-the-art reviews on structural design, performance assessment, classification, and other aspects of structural engineering [3, 38, 76].

Several studies in literature have investigated predicting the seismic target response of the structures using ML algorithms [7, 8, 19, 22, 24, 34, 42, 60]. For instance, Asgarkhani et al. [7] investigated the prediction of maximum interstory drift, maximum residual interstory drift, and maximum roof interstory drift demands using stacked ML algorithm. NTAs and incremental dynamic analysis (IDAs) were performed for 2- to 12-story buckling-restrained brace frames. Additionally, 78 far-field GMs were used for analysis. The study considered various ML models, including random forest (RF) [49], bagging regressor (BR) [13], extra-trees regressor (ETR) [31], artificial neural networks (ANNs) [79], recurrent neural networks (RNNs) [27], gradient boosting machine (GBM) [29], extreme gradient boosting (XGBoost) [15], and the proposed stacked ML algorithm. The analysis results indicated that the proposed ML method could predict seismic response parameters with high accuracy. Recently, Noureldin et al. [60] predicted the seismic performance of low- to mid-rise frame buildings, taking into account soil-structure interaction (SSI). The methodology used in the study considers different ML techniques to achieve highest prediction accuracy for seismic demands. In addition, the framework also considered two different types of structural irregularities: mass and stiffness. Another study conducted by [19, 22], utilized advanced tree-based ML methods to predict the structural response (i.e., maximum drift ratio) of mid-rise RC frames with structural irregularities. The study employed different data preprocessing techniques such as missing data imputation using RF and feature selection using Hill Climbing [70]. According to analysis results, RF constantly performed better than other ML models for all RC frames.

Recently, ML has been evolving at an incredible development, leading to significant theoretical advancements and various applications across various domains [77]. One of the key areas of focus in this field is optimization, which has become a major point of interest for researchers. Optimization plays a crucial role in fine-tuning ML models, and their importance is efforts to develop more effective and efficient methods. As is known, some ML algorithms such as tree-based ML algorithms contain many hyperparameters that strongly affect the model's prediction accuracy [28]. Hence, employing an optimization algorithm to properly adjust these hyperparameters is an important task [11, 21, 35]. Various optimization algorithms have been proposed for this purpose, including particle swarm optimization (PSO) [43], genetic algorithm (GA) [30], grasshopper optimization algorithm (GOA) [73], differential evolution (DE) [75], wolf-bird optimizer (WBO) [10], symbiotic organisms search (SOS) [16], whale optimization algorithm (WOA) [57], and so on. Azizi and Zhou [11] investigated an optimized ML approach for predicting structural seismic response using WBO. In the study, GA, PSO, SOS, and WOA were utilized for comparison. According to analysis results, the WBO algorithm provided higher accuracy prediction than other metaheuristic approach. Another study was conducted by Demir and Sahin [21] employed the PSO algorithm for hyperparameter optimization of the gradient boosting models, specifically PSO-XGBoost, PSO-LightGBM, and PSO-CatBoost. In this study, the analysis results with the PSO optimized models were compared to other models which used default parameters. The results indicated that the PSO algorithm achieved higher prediction accuracy with optimized models compared to those using default parameters models. In addition to the above studies, GOA presented by Saremi et al. [73] was also used to feature selection problems in the literature for the different areas of sciences [4, 40, 41, 51]. Ibrahim et al. [40] studied support vector machines (SVM) by using several optimization algorithms, such as multiverse optimizer (MVO) [58], GOA, the firefly algorithm (FF) [81], GA, and PSO to compare the analysis results. According to analysis results, GOA-SVM showed the higher performance compared to other algorithms. Another study conducted by Kamel and Yaghoubzadeh [41], proposed the feature selection method using GOA for the increase the accuracy of the results. The analysis results indicated that using GOA for feature selection resulted in greater efficiency and increased the SVM algorithm's accuracy by 97.14%.

Despite significant progress in using ML techniques to predict seismic demands for RC frames, the application of these methods, particularly the CatBoost algorithm, remains relatively underexplored [33, 67, 80]. There is a notable gap in the literature regarding the use of the CatBoost algorithm for forecasting maximum drift ratio (*MDR*) specifically for low-rise RC frames. This study seeks to address this gap by conducting an initial investigation into the effectiveness of data-driven ML techniques for predicting *MDR* responses in low-rise RC frames, which have 3-story with regular (R3) and vertically irregular (IR3) frames were studied. The main innovations of the paper can be summarized as follows:

- Investigating the relatively unexplored application of the CatBoost algorithm to predict *MDR* for R3 and IR3 RC frames.
- Introducing and applying a GOA-based optimization framework for CatBoost (CatGrass) to enhance its predictive performance.
- Employing an advanced feature engineering framework to optimize the CatBoost model and to improve its predictive capabilities.
- Enhancing the robustness and reliability of *MDR* predictions by reducing uncertainties in seismic demand forecasting for RC frames.

 https://doi.org/10.1007/s00521-025-11387-z

- Contributing to seismic design methodologies by integrating ML with optimization techniques to offer a more precise and dependable approach to seismic demand assessment.

Moreover, the study considers the following contributions which are typically ignored in research studying problems using ML algorithms in structural engineering domain.

- Prediction performance of the CatGrass model was systematically compared with some popular ML algorithms, such as RF, SVM, and XGBoost to make a more comprehensive evaluation.
- A SHAP (SHapley Additive exPlanations) based feature importance analysis was applied to evaluate and measure the relative importance of input features.
- Computation costs of the CatGrass model and other ML models were evaluated to observe the trade-off between accuracy and computational efficiency.
- A statistical significance test was employed to determine whether there is a statistical significance between the actual and predicted values.

2 Methodology

In this study, an advanced ML pipeline called CatGrass was employed to predict the structural response of regular and vertically irregular 3-story RC frames. The CatBoost algorithm optimized by GOA was utilized for generation ML models using several features. These features consisted of 21 input parameters, comprising 20 different IMs and $S_a(T_1)$, along with a target feature represented by Δ_{max}/H (i.e., *MDR*). This framework incorporated an extensive pipeline, involving data preprocessing techniques such as multivariate outlier detection and replacement (*outForest*) analysis for outlier detection, remove missing values (*na.omit*) function for cleaning rows containing any null entries, feature transformation with Box-Cox transformation, and feature selection through the Chi-square method. Figure 1 presents the ML pipelines for the generated models, providing a visual summary of the processes. The performance measurements (i.e., R^2, MSE, RMSE, MAE, and MAPE) were utilized to evaluate each ML model. Also, a benchmark ML pipeline was built using CatBoost algorithm using no feature engineering process. The main reason for the implementation of two different ML pipelines is to observe the extent to which feature engineering applications affect model prediction performance. It is worth noting that the configuration of the computational power executed for this study is listed in Table 1.

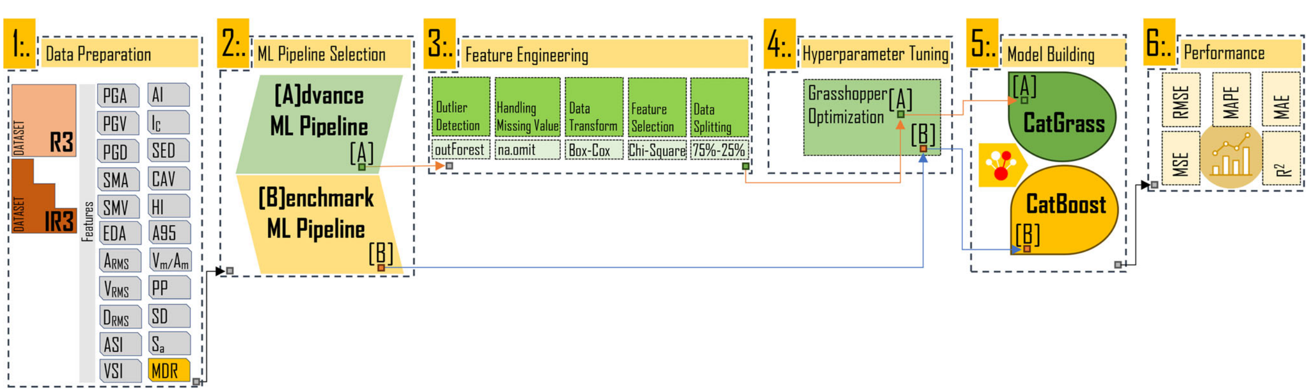


Fig. 1 The framework of [A] and [B] employed for model development and performance evaluation

Table 1 Configuration of the computational power executed during the development of the ML models

Components	Details
RAM	64 GB
Processor	4.0 GHz AMD Ryzen 9 3950X CPU
Operating system	Windows 10

2.1 Details of structure properties and the dataset

2.1.1 Analysis models of low-rise RC frames

The buildings stock in Türkiye and the world mostly consists of low- and medium-rise buildings. Therefore, it is important to obtain the seismic response of these buildings for future earthquakes. In this study, low-rise RC frames of R3 and IR3 were studied. Figure 2 shows these frames which were designed by Hatzigeorgiou and Liolios [36]. Low-rise frames consist of the beam columns. As can be seen from Hatzigeorgiou and Liolios [36], all frames were assumed to be designed on local soil classes B according to EC8.

Figure 2 also presents the geometrical and reinforcement properties of both beams and columns. The floor height is 3.0 m for both low-rise frames. Dimensions of the columns and beams are 30/30 cm and 30/40 cm for both R3 and IR3 frames, respectively. The accepted values for the concrete strength and the strength of reinforcements are 20 MPa and 500 MPa, respectively. Dead and live loads are equal to 20 kN/m and 10 kN/m loading on the beams, respectively. It should be noted that self-weight of beams is not included in dead load. Using the loading (dead, live, and self-weight), reinforcing details, and cross-sectional dimensions, all frames were nonlinear modeled with SAP2000 [72]. Mander model [53] was used for confined and unconfined concrete and stress–strain models with strain hardening for steel. The lumped plastic hinges are considered, and the hinges are assigned ends of the frame's members at the half of the members section height for considered direction. The first mod period of the R3 and IR3 is found to be 0.611 s and 0.470 s.

Capacity curves were obtained using pushover analysis. In this analysis, first mode shape is considered for lateral load patterns and second-order effects which are known as *P-Δ* effect are also considered to obtain capacity curves. The other details, such as assumptions of nonlinear modeling and information about capacity curves, can be found in Demir [18]. The ratio of the base shear to calculated seismic weight of the frames which also known as "lateral strength ratio, *Fy/W*" is calculated for both R3 and IR3 frames as 40% and 59%, respectively.

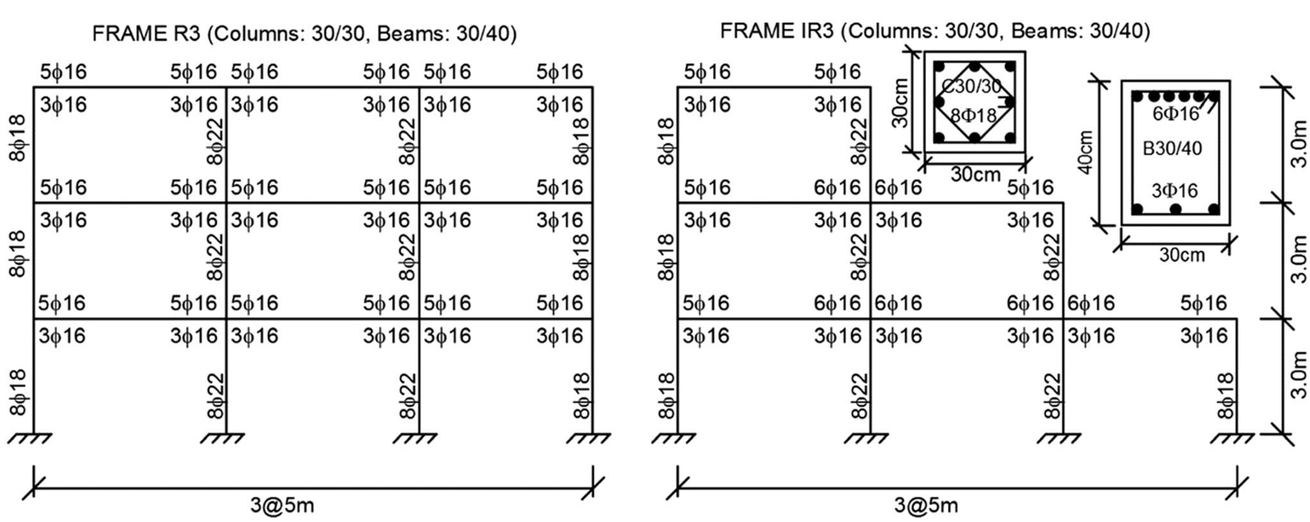


Fig. 2 Properties of the RC frames for R3 and IR3 buildings

2.1.2 Ground motion records and IMs

In EC8, three different GM selection criteria are given for nonlinear dynamic analysis: (1) At least three GMs should be used; (2) the mean of spectral acceleration values of selected GM at the $T = 0.0$ s should be higher than the corresponding target spectrum (T is fundamental period of the structure); and (3) the ratio of the mean of the selected GM spectral acceleration values to the corresponding target spectrum should be higher than the 0.90 for between the 0.2 T and 2.0 T. In addition, mean structural response is used if at least seven GM records are used in the time history analysis. In this study, 2300 GM records were used for calculated *MDR* of the R3 and IR3 frames. All GM records are obtained from Demir [18]. These GM records were downloaded from PEER database (Ancheta et al. [1]). The magnitude, epicentral distances, and shear wave velocity properties of the GM records are $M_w > 5.0$, $10 < R < 60$, and $V_{s,30}$ are between 360 and 800 m/s which compatible with EC8 local soil class B. The period ranges for compatibility between the mean spectra of the selected GM records and corresponding target spectra are assumed between 0.08 s-2.50 s. Finally, scaling GM records is an important task for the nonlinear dynamic analysis [19, 22, 62]. In this study, scaling factors are considered between 0.25 and 4.00 for scaling GMs. The individual spectrums of the 2300 selected GMs (in shadow area), target, mean spectra of selected GMs, and mean $\pm$ 0.5xSD (standard deviation) are given in Fig. 3. It is seen that the compatibility between the mean of the selected GMs and target spectra is high. Therefore, it can be said that the mean spectra of selected GMs meet the GM selection criteria according to EC8 conditions.

In this study, 20 different IMs and S_a values at the first mode periods $S_a(T_1)$ for each record were obtained. Prediction of *MDR* for the R3 and IR3 frames based on these IMs and $S_a(T_1)$ was investigated via ML approaches. It should be noted that there is much research about the relationship between the IMs and the seismic response of the buildings [18, 45, 61]. According to these studies, there is a robust correlation between the seismic demand and the IMs. For this study, all IMs are presented in Fig. 4.

IMs show the amplitude, duration, and frequency content of the GM records. In these IMs, six amplitude content (peak ground acceleration (*PGA*), peak ground velocity (*PGV*), peak ground displacement (*PGD*), sustained maximum acceleration (*SMA*), sustained maximum velocity (*SMV*), and effective design acceleration (*EDA*)) and five different amplitude and frequency content (root mean square of acceleration (A_{RMS}), root mean square of velocity (V_{RMS}), root mean square of displacement (D_{RMS}), acceleration spectrum intensity (*ASI*), and velocity spectrum intensity (*VSI*)) were used. Furthermore, some IMs describe amplitude, frequency, and/or duration. These are V_{max}/A_{max}, predominant period (*PP*), and significant duration (*SD*). Lastly, the IMs with amplitude, frequency, and duration contents, such as Arias intensity (*AI*), characteristic intensity (*CI*), specific energy density (*SED*), cumulative absolute velocity (*CAV*), Housner Intensity (*HI*), and level of the acceleration which contains up to 95% of the *AI* (*A95*) were also used in the study. $V_{\max}/A_{\max}$ and *PP* represent frequency of the records. $Sa(T_1)$ values of R3 and IR3 frames are given in Fig. 5.

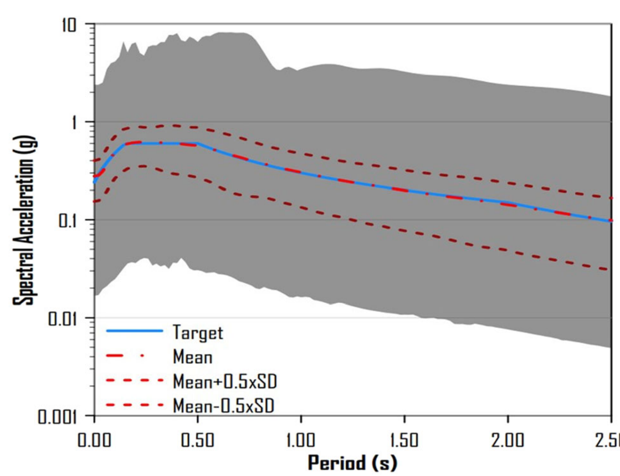


Fig. 3 Target and mean spectra for 2300 ground motion records

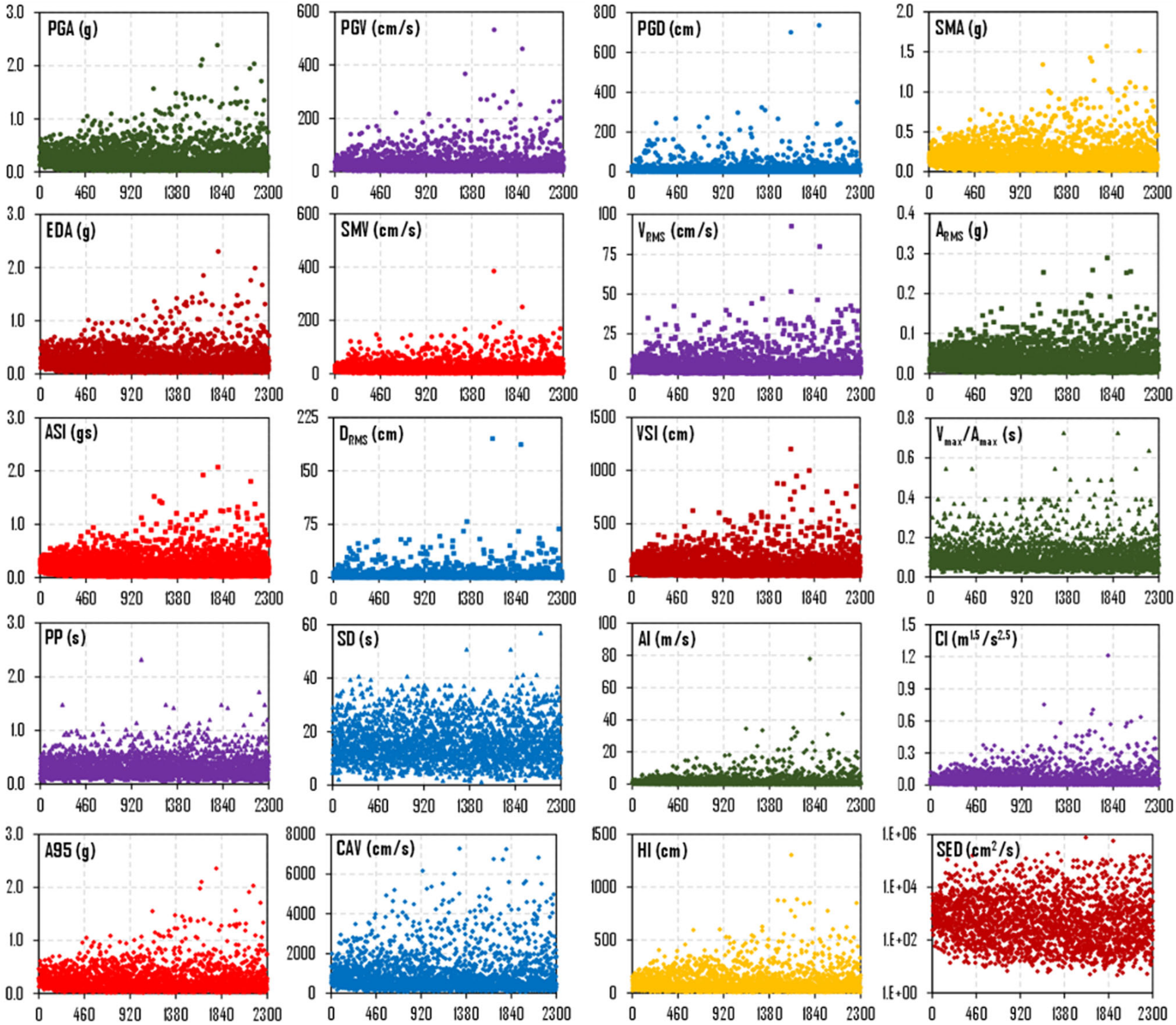


Fig. 4 IMs used in the study

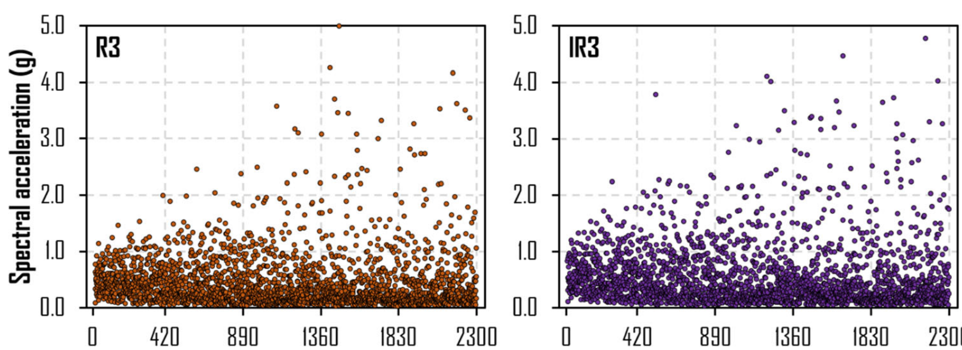


Fig. 5 Spectral acceleration values at the first periods of frames ($S_a(T_1)$)

Table 2 Statistical parameters for 21 features used in the study

Feature	Unit	Mean	Maximum	Minimum	CoV
PGA	g	0.277	2.391	0.017	0.890
PGV	cm/s	30.373	533.446	1.41	1.262
PGD	cm	15.571	736.47	0.311	2.411
V_{max}/A_{max}	s	0.112	0.726	0.019	0.647
A_{RMS}	g	0.035	0.29	0.002	0.879
V_{RMS}	cm/s	5.268	92.678	0.161	1.296
D_{RMS}	cm/s	3.731	195.402	0.032	2.491
AI	m/s	1.818	77.78	0.006	2.156
CI	$m^{1.5}/s^{2.5}$	0.059	1.21	0.001	1.403
SED	cm^2/s	5226.463	772,988.803	4.784	4.841
CAV	cm/s	987.640	7285.611	64.035	0.989
ASI	gs	0.237	2.073	0.016	0.885
VSI	cm	109.954	1203.563	8.129	1.042
HI	cm	101.107	1306.043	7.55	1.097
SMA	g	0.205	1.573	0.015	0.852
SMV	cm/s	21.690	384.89	1.32	1.195
EDA	g	0.262	2.306	0.014	0.889
A95	g	0.273	2.361	0.016	0.892
PP	s	0.338	2.32	0.06	0.616
SD	s	16.083	56.77	1.19	0.489
S_a (R3)	g	0.486	3.638	0.014	1.147
S_a (IR3)	g	0.574	3.483	0.016	1.045

Moreover, the statistical parameters of the IMs and $S_a(T_1)$ values, including mean, maximum and minimum values, and *CoV* of R3 and IR3 frames are given in Table 2. It is seen that the *CoV* values of the IMs are generally high and range between 0.489 and 4.841 for *SD* and *SED*, respectively.

2.1.3 Analysis results and data analysis

In this study, 4600 (2300 GM records for R3 and IR3 frames) NTH analyses were performed. As a result of the NTH analysis, the maximum drift (Δ_{max}) value was calculated for each GM record and both R3 and IR3 frames. Then, the *MDR* (Δ_{max}/H) was obtained by dividing the maximum drift value by the frame's height (height is 9.0 m). Δ_{max}/H values of each R3 and IR3 RC frames for 2300 GM records are given in Fig. 6. The maximum and minimum Δ_{max}/H values are computed as 10.40–0.037% and 8.43–0.03% for R3 and IR3, respectively. The mean of the 2300 Δ_{max}/H values are found to be 0.598% for R3 and 0.500% for IR3. In addition, the $CoV_{(\Delta/H)}$ values are calculated as 1.261 and 1.158 for R3 and IR3, respectively. Figure 6 shows that the dispersion of the Δ_{max}/H values is considerable high. This issue was not considered in the study when selecting the GM record and just the

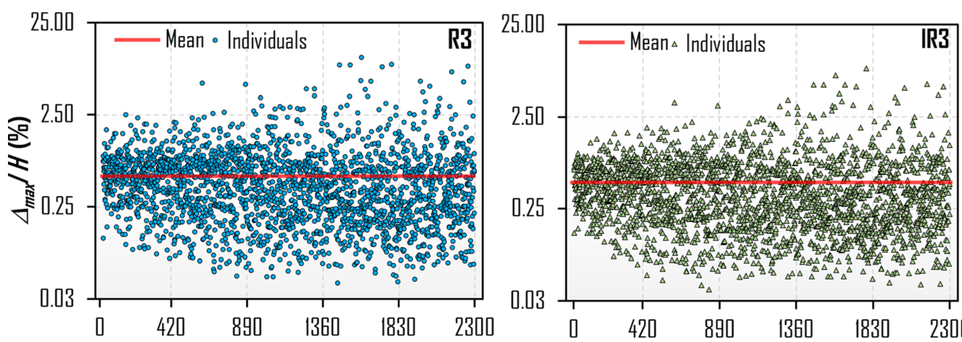


Fig. 6 Δ_{max}/H values of RC frames for 2300 ground motion records

compatibility between the mean spectrum and the target spectra was checked. The compatibility between the individual spectrum and the target spectrum was not considered in the study because of only EC8 GM record selection considered. On the other hand, there is a strong correlation between the dispersion of individual spectrum and dispersion of seismic response ([5], [23]). However, dispersion of seismic response was not considered and also was not the purpose of this study.

2.2 Feature engineering framework

In this study, a specialized feature engineering framework (advanced ML pipeline) was employed on the studied dataset to ensure the robustness and accuracy of the predictive models. The first step involved outlier detection using the *outForest* approach [9]. The implementation of the outlier forest in the *R* library was preferred for identifying and removing anomalous data points from the dataset. *outForest* identifies outliers by regressing each numeric variable against all other variables using a random forest model. If the scaled absolute difference between the observed value and the out-of-bag prediction from the random forest is unusually large, the value is flagged as an outlier [54]. After removing outliers in the dataset, it was cleaned of any missing values (removed outliers) using the *na.omit* function, which systematically removes rows containing any null entries.

To address skewness in the distribution of features, the Box-Cox transformation was applied. The Box-Cox transformation [12] is a powerful tool for normalizing non-normal data, thereby stabilizing variance and making the data more amenable to predictive modeling. Mathematically, for a given feature y with skewed distribution, the Box-Cox transformation is defined as:

$$y^{\lambda} = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda} & (\lambda \neq 0) \\ \log y & (\lambda = 0) \end{cases} \tag{1}$$

where λ is the transformation parameter that optimizes normality. By applying the Box-Cox transformation, the features are made more symmetric. An example of R3 and IR3 datasets in the case of before and after Box-Cox transformation is given in Table 3. Upon evaluating the skewness of features in the raw data for R3 and IR3, it is found that the skewness values for nearly all features, except *PP* and *SD*, exceed 1.0, indicating a high degree of skewness in their distribution. However, after applying the Box-Cox transformation, it is observed that the distributions of all features become symmetric.

Feature selection was conducted using the Chi-square test, a statistical method employed to evaluate the independence between features and the target variable. The Chi-square test (Li et al. [48]) is a widely utilized method for feature selection, and it is useful for identifying relevant features, reducing the dimensionality and improving model performance. This statistical technique aims to identify features that exhibit significant dependency on the target variable. The Chi-square test assesses the relationship between two variables by comparing observed and expected frequencies, thereby determining whether they are related or independent [74].

In cases where the variables are dependent, the observed frequencies will substantially deviate from the expected frequencies, yielding a high Chi-square value. Conversely, if the variables are independent, the observed frequencies will closely align with the expected frequencies, resulting in a low Chi-square value. A higher Chi-square value indicates a stronger dependency between the variables, making the feature more suitable for inclusion in model training. In contrast, a lower Chi-square value suggests independence, implying that the feature may not contribute significantly to the model and, therefore, may be excluded from further analysis. Table 4 presents the features identified by the Chi-square feature selection method from each dataset. It is seen that the features of V_{max}/A_{max}, *PP*, *SD*, D_{RMS}, *PGD* and V_{max}/A_{max}, *SD*, D_{RMS} were extracted from the dataset R3 and IR3, respectively.

Table 3 Variation of skewness values and distribution types of the features after Box-Cox transformation

Feature	R3				IR3			
	RAW		Box-Cox		RAW		Box-Cox	
	SV	DT	SV	DT	SV	DT	SV	DT
PGA	1.66	Highly skewed	0.01	Symmetric	1.66	Highly skewed	0.00	Symmetric
PGV	4.27	Highly skewed	0.04	Symmetric	4.19	Highly skewed	0.06	Symmetric
PGD	5.07	Highly skewed	0.25	Symmetric	5.11	Highly skewed	0.29	Symmetric
V_{max}/A_{max}	1.72	Highly skewed	0.09	Symmetric	1.69	Highly skewed	0.10	Symmetric
A_{RMS}	1.45	Highly skewed	0.04	Symmetric	1.45	Highly skewed	0.05	Symmetric
V_{RMS}	2.88	Highly skewed	0.00	Symmetric	3.03	Highly skewed	0.03	Symmetric
D_{RMS}	5.08	Highly skewed	0.15	Symmetric	5.06	Highly skewed	0.19	Symmetric
AI	5.96	Highly skewed	− 0.23	Symmetric	5.86	Highly skewed	−0.22	Symmetric
CI	3.12	Highly skewed	− 0.27	Symmetric	3.15	Highly skewed	−0.26	Symmetric
SED	10.47	Highly skewed	0.19	Symmetric	9.55	Highly skewed	0.23	Symmetric
CAV	1.89	Highly skewed	− 0.06	Symmetric	2.06	Highly skewed	−0.03	Symmetric
ASI	1.40	Highly skewed	0.02	Symmetric	1.39	Highly skewed	0.01	Symmetric
VSI	3.16	Highly skewed	− 0.01	symmetric	3.16	Highly skewed	0.00	Symmetric
HI	3.46	Highly skewed	0.05	Symmetric	3.44	Highly skewed	0.06	Symmetric
SMA	1.38	Highly skewed	− 0.04	Symmetric	1.39	Highly skewed	−0.04	Symmetric
SMV	2.80	Highly skewed	0.04	Symmetric	2.96	Highly skewed	0.07	Symmetric
EDA	1.70	Highly skewed	0.03	Symmetric	1.70	Highly skewed	0.02	Symmetric
A95	1.66	Highly skewed	0.01	Symmetric	1.66	Highly skewed	0.00	Symmetric
PP	0.91	Moderately skewed	0.03	Symmetric	0.92	Moderately skewed	0.03	Symmetric
SD	0.61	Moderately skewed	− 0.08	Symmetric	0.62	Moderately skewed	−0.06	Symmetric
S_a	2.57	Highly skewed	− 0.11	Symmetric	2.01	Highly skewed	−0.11	Symmetric

*SV: skewness value [17], DT: distribution type

Table 4 Selected features after the Chi-square test

Dataset	Number of features	Features selected by Chi-square
R3	16	S_a, VSI, PGV, HI, SMV, AI, CI, EDA, SMA, PGA, ASI, A95, V_{RMS}, A_{RMS}, CAV, SED
IR3	18	S_a, VSI, PGV, HI, SMV, AI, CI, EDA, SMA, PGA, ASI, A95, V_{RMS}, A_{RMS}, CAV, SED, PGD, PP

Subsequently, the dataset was split into training and test sets in a 75–25% ratio. This step ensures that the models are trained on a substantial portion of the data while reserving a sufficient subset for validation purposes. The choice of a 75–25% split balances the need for ample training data with the necessity of a robust testing phase, thus mitigating overfitting.

The final stage of the preprocessing pipeline involved integrating the selected features into a predictive model built using the CatBoost algorithm, optimized through GOA. CatBoost, a gradient boosting algorithm that excels in handling categorical data, was chosen for its ability to mitigate overfitting and deliver high accuracy [21]. The integration of GOA into CatBoost enhances the model's performance by optimizing hyperparameters such as learning rate, depth, and iterations. GOA, inspired by the swarming behavior of grasshoppers, operates by iteratively adjusting the hyperparameters to minimize the objective function, thereby improving the predictive power of the model. Details of these algorithms are presented in the following section.

2.3 Details of optimization and machine learning algorithms

2.3.1 Grasshopper optimization algorithm (GOA)

GOA is a highly effective population-based metaheuristic algorithm that expertly models the behavior of grasshopper swarms in nature to solve optimization problems [73]. The algorithm operates by simulating both repulsion and attraction forces among virtual grasshoppers within a search space. Repulsion forces enable grasshoppers to explore widely across the search space, while attraction forces drive them toward promising areas for exploitation [37]. This dual approach also makes the GOA effective in avoiding the problem of stagnation at local optima, which is a common issue in optimization problems. In GOA, the grasshoppers collectively form a network, creating a community where each individual's position is synchronized with the others. This interconnected network allows for the direction of foraging to be determined based on the positions of other individuals in the group [25]. The following equation is the mathematical model used to confidently simulate the swarming behavior of grasshoppers:

$$X_i = S_i + G_i + A_i \tag{2}$$

where X_i determines the position of the *i-th* grasshopper, S_i and G_i are the social interaction between grasshoppers and the gravity force on the *i-th* grasshopper, respectively, and A_i is the wind advection. Moreover, random behavior can be written using random numbers with the following equation:

$$X_i = r_1 S_i + r_2 G_i + r_3 A_i \tag{3}$$

where r_1, r_2, and r_3 are the random numbers in the range [0–1]. The social interaction S_i is defined as follows:

$$S_i = \sum_{j=1, j\neq i}^{N} s(d_{ij})\hat{d}_{ij} \tag{4}$$

In which, d_{ij} is the distance between the *i-th* and *j-th* grasshopper and can be determined as $d_{ij} = |x_j - x_i|$. $\hat{d}_{ij}$ is the unit vector between the *i-th* and *j-th* grasshoppers represented by $\hat{d}_{ij} = \frac{(x_j - x_i)}{d_{ij}}$. The s function is the social forces and can be determined in Eq. (5).

$$s(r) = fe^{-r/l} - e^{-r} \tag{5}$$

where f denotes the attraction intensity, l shows the attractive length scale, and r is the random number.

The G_i and A_i parameters given in Eq. (2) can be computed using the following equations:

$$G_i = -g\hat{e}_g \tag{6}$$

$$A_i = u\hat{e}_w \tag{7}$$

The g, u, $\hat{e}_g$, and $\hat{e}_w$ denote gravitational constant, constant drift, unity vector toward the center of the earth, and unity vector toward the direction of wind, respectively.

Equation (2) can be generalized by substituting S_i,G_i, and A_i as below:

$$X_i^d = c\left(\sum_{j=1,j\neq i}^{N} c\frac{ub_d - lb_d}{2} s\left(\left|x_j^d - x_i^d\right|\right)\frac{x_j - x_i}{d_{ij}}\right) + \hat{T}d \tag{8}$$

where N shows the grasshoppers' number, ub_d and lb_d are the *d-th* dimension of upper boundary and lower boundary, $\hat{T}d$ is the *d-th* dimension value in the best solution so far obtained, and c is a decreasing coefficient which is used to balances the entire swarm's exploration and exploitation of the target according to the number of iterations through the following equation:

$$c = c_{\max} - l\frac{c_{\max} - c_{\min}}{L} \tag{9}$$

where the maximum and minimum values are represented by c_{max} and c_{min}, l corresponds to the current iteration, and L denotes the highest iteration number.

2.3.2 Categorical boosting (CatBoost)

CatBoost is a kind of gradient boosting decision tree algorithm (GBDT) that stands out due to its superior performance compared to the other publicly available boosting algorithms [65]. It is a useful algorithm for solving problems in areas such as classification, regression, and ranking. CatBoost has an innovative approach to dealing with categorical variables. Unlike many other gradient boosting algorithms that require preprocessing such as one-hot encoding or label encoding, CatBoost implements an algorithm known as ordered boosting. This technique allows CatBoost to process categorical features directly by sorting their values and creating splits based on this sorted order during training [21]. CatBoost incorporates a unique feature known as a "prior term" which is designed to augment the performance of greedy target statistics as given in Eq. 10. The introduction of this prior term fundamentally alters the way the algorithm functions, enabling it to generate more accurate models. Furthermore, this integration is not merely for enhancing performance, it plays a crucial role in mitigating the risk of overfitting (Lui and Setiono [50]).

$$x_{\sigma_{p,k}} = \frac{\sum_{j=1}^{p-1}\left[x_{\sigma_{j,k}} = x_{\sigma_{p,k}}\right]Y_{\sigma_j} + \alpha\cdot P}{\sum_{j=1}^{p-1}\left[x_{\sigma_{j,k}} = x_{\sigma_{p,k}}\right] + \alpha} \tag{10}$$

where P represents the prior term and α indicates the weight coefficient greater than 0.

2.4 Hyperparameter tuning and optimization

Hyperparameter tuning plays a pivotal role in optimizing the performance of ML models by directly influencing the behavior of training algorithms [82]. Several common hyperparameter tuning approaches have limitations such as increased search space complexity, high computational cost, and elevated variance [21]. GOA excels at balancing exploration and exploitation, allowing it to efficiently converge toward globally optimal solutions in high-dimensional search spaces. In this study, GOA was utilized to select the optimal hyperparameters for the CatBoost algorithm (Table 5). GOA is governed by a set of control parameters, including fitness criteria, type of optimization (*optimType*), number of variables (*numVar*), maximum number of iterations (*maxIter*), population size (*numPopulation*), and a matrix containing the range of variables (*rangeVar*). A full list of GOA parameters and their default settings can be found in the GOA documentation [68]. Table 6 outlines the GOA control parameters and tuned hyperparameters of the CatBoost algorithm.

The construction of the CatBoost model commenced with identifying optimal hyperparameters by applying the GOA algorithm. The most critical and influential hyperparameters, *depth*, *learning_rate*, *l2_leaf_reg*,

Table 5 CatBoost hyperparameters and their ranges

No	Hyperparameters	Explanation	Range
1	Depth	Depth of the tree	1.0–10
2	Learning_rate	Boosting learning rate	0.01–1.0
3	l2_leaf_reg	Coefficient at the L2 regularization term of the cost function	2.0–10
4	Random_strength	The level of randomness	0.0–10
5	Subsample	Ratio of training instances	0.01–1.0
6	Model_shrink_rate	The constant used for determining the coefficient to multiple the model on each iteration	0.0–1.0

Table 6 Hyperparameters for the CatBoost algorithm with GOA control parameters

Model	Hyperparameters					
	Depth	Learning rate	l2_leaf_reg	Random_strength	Subsample	Model_shrink_rate
R3	7.110	0.672	6.862	0.485	0.428	0.0007
IR3	10.00	0.713	3.534	0.519	0.696	0.0002
Model	GOA control parameters					
	FC*	Maxiter	Rangevar	Numpopulation	NumVar	Optimtype
R3	RMSE	100	Matrix (2 × 6)	20	6	MIN
IR3	RMSE	500	Matrix (2 × 6)	40	6	MIN

*FC: fitness criteria

random_strength, *subsample*, and *model_shrink_rate* were utilized for optimization by the algorithm. After completion of the GOA optimization process, the optimal values for the selected hyperparameters were determined for the R3 and IR3 cases, given in Table 6.

2.5 Evaluation criteria for algorithm performance

Understanding and addressing errors is an essential aspect of evaluating the accuracy of an ML model. The discrepancy or deviation that appears between the actual data (y_i) and the predicted data ($\hat{y}_i$) is referred to as the "error." This error serves as a measure of the accuracy of the ML model's predictions. If this error falls within acceptable boundaries, it signifies that the model is performing well and is ready to be put into use. However, if the error exceeds the tolerable limits, it indicates that the model's predictions are not accurate enough. There are numerous methods available to estimate the accuracy of an ML model's performance. Some of the most popular and widely used methods include the mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and coefficient of determination (R^2) (Table 7). Each of these methods provides a different perspective on the model's accuracy, allowing data scientists to gain a comprehensive understanding of the model's performance and make necessary adjustments. The metrics to be used to evaluate the performance of the algorithms can be given as listed in Table 7 with n which represents test sample size and $\overline{y}_i$ shows mean value.

 https://doi.org/10.1007/s00521-025-11387-z

Table 7 Performance evaluation metrics for regression models

Name of evaluation metric	Equation
Mean squared error	$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
Root mean squared error	$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$
Mean absolute error	$MAE = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$
Mean absolute percentage error	$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert$
Coefficient of determination	$R^2 = 1 - \frac{\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert^2}{\sum_{i=1}^{n}\lvert y_i - \bar{y}_i\rvert^2}$

3 Results and discussion

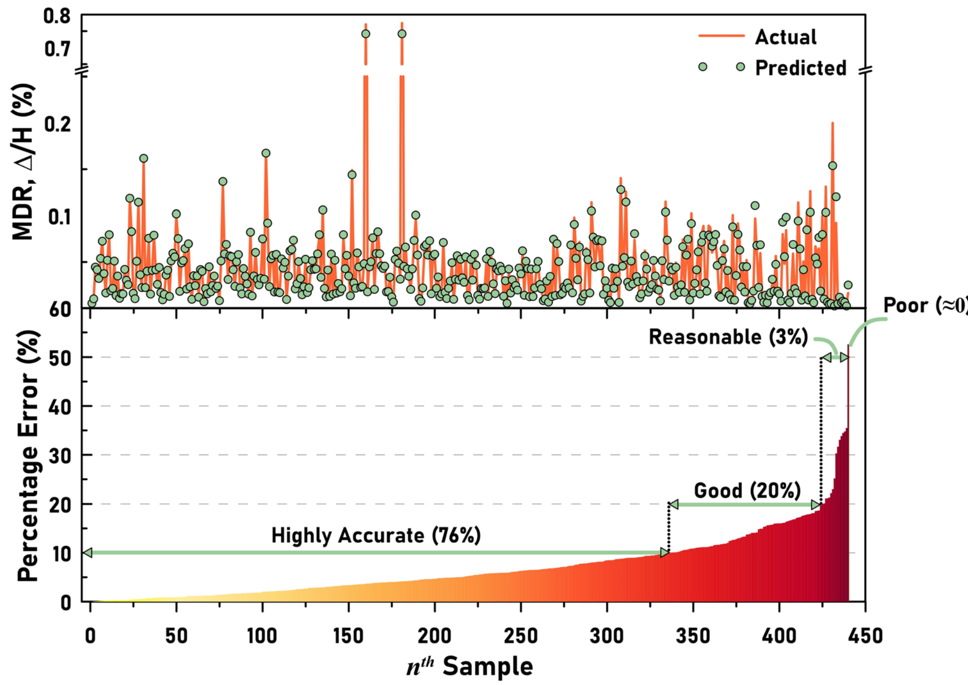
3.1 Evaluation of R3

In this paper, the test data are utilized to rigorously evaluate the performance of ML models through a range of statistical metrics. Each of these metrics offers unique insights into the predictive capacity and accuracy of the model. In particular, lower values of RMSE, MSE, and MAE signify enhanced model reliability and statistical robustness, suggesting that the model's predictions are more aligned with actual outcomes. Meanwhile, the R^2 value, which varies between 0 and 1, measures the proportion of the variance in the dependent variable explained by the independent variables. Generally, R^2 values above 0.7 or 0.8 are regarded as indicative of a strong model fit, reflecting a higher level of explanatory power. Another critical metric in evaluating model performance is MAPE, which provides an easily interpretable measure of prediction accuracy. MAPE expresses the average deviation between predicted and actual values as a percentage, offering a more intuitive understanding of the model's performance. A MAPE value below 10% is generally classified as *"highly accurate,"* representing an excellent model. Values between 10 and 20% are considered *"good,"* while those between 20 and 50% indicate *"reasonable"* predictive accuracy. However, a MAPE value exceeding 50% points to a model with *"poor"* performance, suggesting that its predictions deviate substantially from actual values and may not be reliable for practical applications [47]. Table 8 summarizes the overall performance metric results for R3 using the CatGrass framework. As can be seen from the table, all metrics have excellent scores, indicating a highly accurate prediction performance. Moreover, Fig. 7 thoroughly illustrates the comparison between the actual and predicted values of each test sample. The comparison between the actual and predicted values clearly shows a strong alignment between them. An individual analysis of the test data reveals that 76% of the test data were predicted with "*highly accurate,*" indicating a minimal percentage difference between the actual and predicted values. Furthermore, 20% of the predictions were classified as *"good"* while the proportion deemed "*reasonable*" accounts for 3%. When averaged across the dataset, as indicated in Table 8, MAPE was calculated to be 7%, indicating a highly effective prediction performance.

Table 8 Performance results of the ML model for R3

Evaluation metrics				
R^2	MSE	RMSE	MAE	MAPE (%)
0.9928	0.00002	0.0053	0.0028	7.0074

Fig. 7 Comparison of actual and predicted Δ_{max}/H values with percentage error for R3



3.2 Evaluation of IR3

Table 9 indicates the performance metrics results for IR3. Similar to R3, IR3 results exhibited highly successful performance outcomes. This clearly demonstrates the effectiveness and robustness of the applied CatGrass approach, highlighting its ability to deliver reliable results in the context of the given analysis or experiment. Furthermore, for IR3, a detailed comparison of the actual and predicted values for each test data is presented in Fig. 8. The analysis reveals that 70% of the test data were predicted with *"highly accurate."* Additionally, 21% of the samples were classified as *"good"* predictions, 8% of the samples were predicted with *"reasonable"* accuracy, and finally, a small fraction of 2% were categorized as *"poor"* predictions, highlighting areas for potential improvement. Although the results obtained for IR3 are somewhat behind compared to those achieved for R3, both RC cases have still yielded exceptionally high *MDR* predictions. This indicates that despite some differences in performance, the overall predictive accuracy for *MDR* remains notably strong across both scenarios.

3.3 Effectiveness of the CatGrass Model as compared to CatBoost Model

To assess the effectiveness of the CatGrass model, a benchmark model was built using the same dataset and the results of this comparison are illustrated in Fig. 9. It should be noted that no feature engineering was applied to the raw model, it was only optimized with GOA. It is clearly seen that the proposed CatGrass model significantly outperforms the benchmark model in terms of prediction performance. A detailed examination of the R^2 values reveals that CatBoost model yielded R^2 values of 0.86 and 0.89 for R3 and IR3, respectively. However, by employing the novel framework to CatBoost (i.e., CatGrass), an improvement of 11% to 15% was observed in R^2 values, reaching up to 0.99. This substantial enhancement underscores the superiority of the proposed method.

Table 9 Performance results of the ML model for IR3

Evaluation metrics				
R^2	MSE	RMSE	MAE	MAPE (%)
0.9922	0.00001	0.0036	0.0024	9.6924

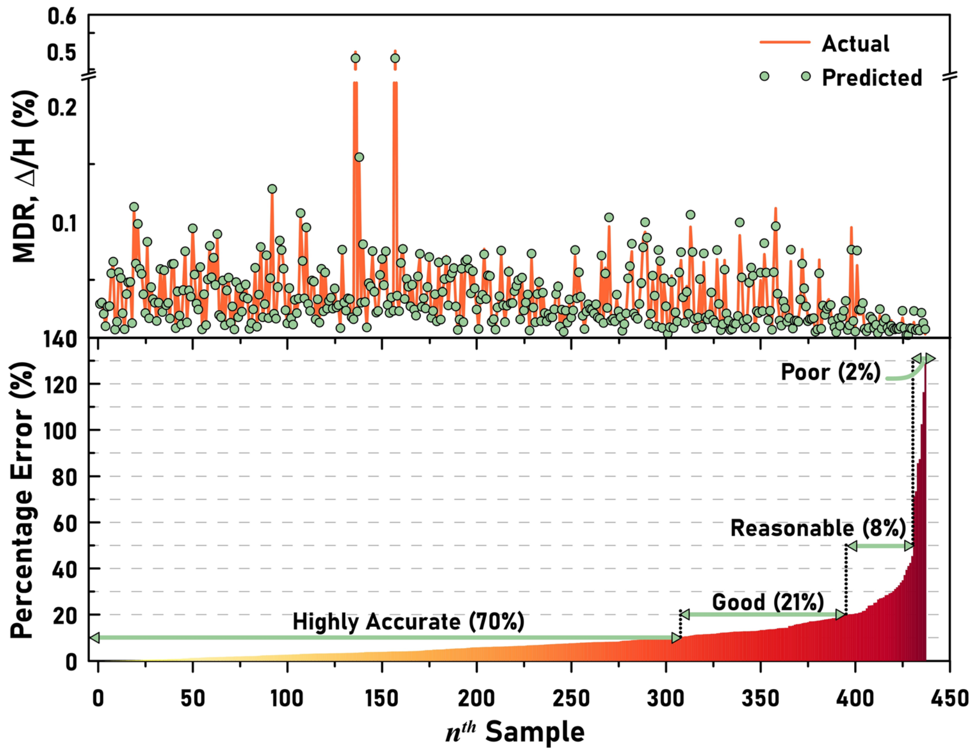


Fig. 8 Comparison of actual and predicted Δ_{max}/H values with percentage error for IR3

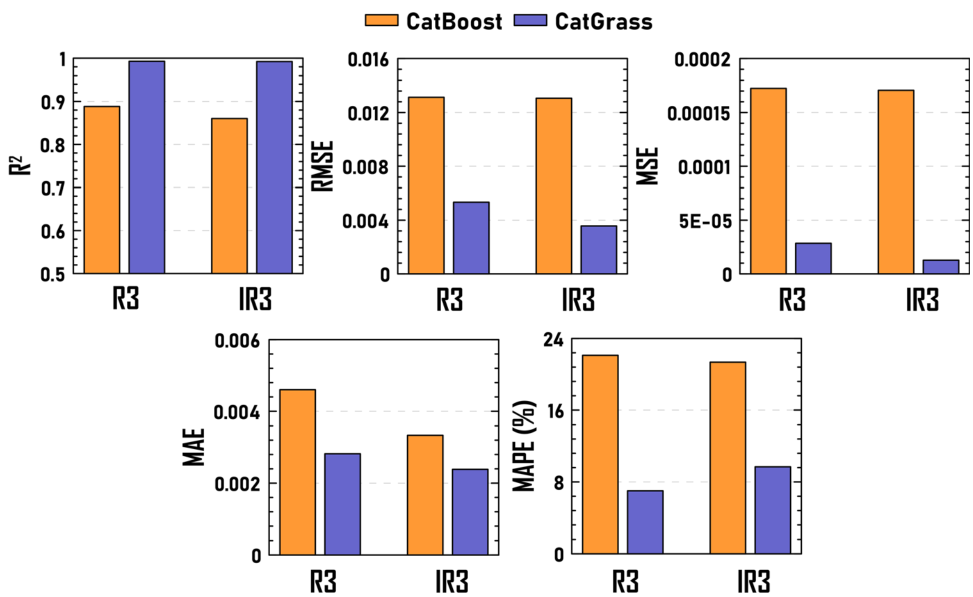


Fig. 9 Performance metric results of the CatBoost and CatGrass models

Moreover, notable improvements were also observed in other key metrics, including RMSE, MSE, MAE, and MAPE, all of which experienced significant decreases. The substantial improvements in all metrics affirm that the CatGrass approach proposed in this study offers a robust and reliable *MDR* prediction model for low-rise RC frames. These findings clearly demonstrate the effectiveness of the integrated feature engineering techniques and optimization strategy, contributing to the development of more accurate and efficient predictive models in structural engineering applications.

3.4 Effectiveness of the CatGrass model as compared to popular ML algorithms

Table 10 compares the prediction results of CatGrass and other popular ML algorithms in terms of R^2, RMSE, MSE, MAE, MAPE, and computational time (s), with the best values highlighted in bold. Table 10 shows that the proposed CatGrass model demonstrates superior predictive performance compared to the other ML models, namely CatBoost, XGBoost, RF, and SVM both for R3 and IR3. CatGrass exhibits the highest R^2 values (0.9928 for R3 and 0.9922 for IR3), and the lowest RMSE, MSE, and MAE values, signifying its superior accuracy in predictive modeling. However, while CatGrass achieves relatively low MAPE values, it is important to highlight that RF for R3 and XGBoost for IR3 outperform CatGrass in this metric. Moreover, it is also evident that CatGrass incurs a significantly higher computational cost compared to the other models. The computational time required for CatGrass is 188.98 seconds for R3 and 190.68 seconds for IR3, which is substantially longer than that of SVM, the most computationally efficient model, with processing times of only 3.06 seconds for R3 and 2.36 seconds for IR3. This observation suggests a trade-off between accuracy and computational efficiency, which must be carefully considered when selecting an optimal model for real-world applications.

3.5 SHAP-based feature importance analysis

The XGBoost algorithm combined with SHAP (SHapley Additive exPlanations) was applied to evaluate and measure the relative importance of input features, as illustrated in Fig. 10. The figure depicts the distribution of SHAP values across individual data points, revealing how different input variables influence the results of prediction algorithms. Dots positioned to the right of the zero line indicate an increasing contribution to the model's prediction, while dots on the left side suggest a decreasing effect. Within the color-coded representation, the red data points demonstrate high feature values compared to their blue counterparts (low feature values). The SHAP summary plots for R3 (a) and IR3 (b) reveal that S_a is the most influential feature in both cases, exhibiting the highest SHAP value. The distribution of SHAP values for S_a in R3 indicates that both high and low values contribute to both positive and negative model outputs without a clear separation. Conversely, high values of S_a are more aligned with positive SHAP values, suggesting a stronger positive impact on the prediction results for

Table 10 Comparison of performance results of the CatGrass model and the other ML models

R3					
	CatGrass	CatBoost	XGBoost	RF	SVM
R^2	**0.9928**	0.8882	0.9629	0.9461	0.7924
RMSE	**0.0053**	0.0131	0.0105	0.0126	0.0252
MSE	**0.0000**	0.0001	0.0001	0.0001	0.0006
MAE	**0.0028**	0.0046	0.0048	0.0048	0.0084
MAPE (%)	7.007	22.115	8.3708	**6.7137**	21.153
Computational time (s)	188.98	104.76	128.76	158.22	**3.06**
IR3					
	CatGrass	CatBoost	XGBoost	RF	SVM
R^2	**0.9922**	0.8602	0.9150	0.9495	0.6133
RMSE	**0.0035**	0.0130	0.0152	0.0118	0.0331
MSE	**0.0000**	0.0001	0.0002	0.0001	0.0011
MAE	**0.0024**	0.0033	0.0044	0.0040	0.0083
MAPE (%)	9.6923	21.350	**6.8376**	6.8447	19.214
Computational time (s)	190.68	99.71	122.04	153.88	**2.36**

*Best values highlighted in bold

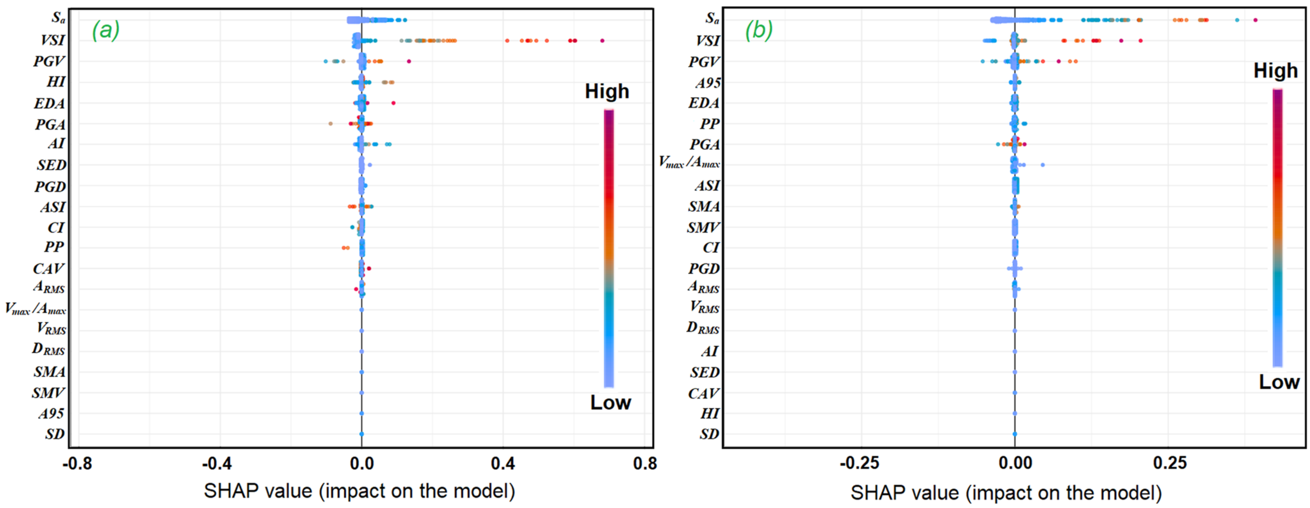


Fig. 10 SHAP importance values of features for R3 (**a**) and IR3 (**b**) cases

IR3. Besides S_a, features such as *VSI*, *PGV*, and *HI* also play significant roles in R3, whereas *A95* gains more importance while *HI* becomes less influential in IR3. These differences indicate that while S_a remains the dominant feature in both cases, the relative contributions of secondary features vary, influencing the model's predictions differently across R3 and IR3.

3.6 Statistical significance analysis

In the evaluation of regression-based ML models, assessing the statistical significance of the relationship between actual and predicted values is crucial for determining model performance. In this section, it was aimed to determine the strength and statistical significance of the linear relationship between actual and predicted values in each dataset. Therefore, the Pearson correlation technique was used to assess the statistical significance of the relationship between predicted and actual values. The Pearson correlation coefficient (denoted as r) is a widely used statistic that measures the strength and direction of a linear relationship between two continuous variables. It ranges from -1 to $+1$, where $r = +1$ indicates a perfect positive linear correlation, meaning that as one variable increases, the other increases proportionally, $r = -1$ signifies a perfect negative linear correlation, whereas one variable increases, the other decreases proportionally, and $r = 0$ denotes no linear correlation between the variables. Besides the correlation coefficient, the *p-value* is a critical output of the Pearson correlation test. A low *p-value* (typically below a significance level of 0.05) suggests that the observed correlation is statistically significant and unlikely to have occurred by random chance.

Based on the prediction results of CatGrass, R3 yielded a Pearson correlation coefficient of $r = 0.979$. This exceptionally high positive value indicates a very strong positive linear relationship between the actual and predicted values. Furthermore, the associated *p-value* was found to be $p < 0.001$. This extremely low *p-value* is far below the conventional significance level of 0.05. For IR3, the Pearson correlation analysis resulted in a correlation coefficient of $r = 0.947$. While slightly lower than that observed for R3, this value still signifies a very strong positive linear relationship between the actual and predicted values. The p-value associated with this correlation was also found to be $p < 0.001$. Statistical significance analysis strongly suggests that the predicted values are statistically significantly related to the actual values. The extremely low *p-values* across all tests indicate that these results are highly unlikely to have occurred by random chance if there were no true relationship between the actual and predicted values. This suggests that the CatGrass model is performing well in predicting the actual values.

4 Conclusions

In this study, an advanced feature engineering pipeline with GOA-optimized CatBoost model was employed to predict the maximum drift ratio of regular and irregular low-rise RC frames. Two distinct RC frames were considered for this purpose: a 3-story regular RC frame (R3) and a vertically irregular RC frame (IR3). 20 different intensity measures (IMs), spectral acceleration (S_a) values at the first mode periods $S_a(T_1)$, and nonlinear time history analyses results for each record are considered as a dataset. The framework used in the study incorporated an extensive workflow, including the data preprocessing techniques such as *outForest* analysis for outlier detection, *na.omit* function for cleaning rows containing any null entries, feature transformation with Box-Cox transformation, and feature selection through the Chi-square method. The hyperparameters of the ML models were optimized using GOA. For comparison, the predictive performance of the GOA-optimized CatGrass models (i.e., CatGrass) was assessed against that of the raw CatBoost model. The findings of this study can be summarized as follows:

- A total of 16 out of 21 features were identified for R3, and 18 features were selected IR3 using the Chi-square feature selection method. Additionally, the features $V_{\max}/A_{\max}$, *PP*, *SD*, D_{RMS}, and *PGD* were eliminated from the R3 dataset, while $V_{\max}/A_{\max}$, *SD*, D_{RMS} were extracted from the IR3 dataset.
- The novel CatGrass model demonstrated exceptionally high performance in predicting *MDR* of low-rise RC frames. The model achieved an R^2 of 0.99, RMSE of 0.0053, a MSE of 0.00002, MAE of 0.0028, and MAPE of 7% for R3. Similarly, the model also reached an R^2 of 0.99, with an RMSE of 0.0036, an MSE of 0.0001, an MAE of 0.0024, and an MAPE of 9.7% for IR3. This study approved the effectiveness of the proposed CatGrass model. It can be considered an efficient tool for seismic design and assessment of RC frames.
- To better demonstrate the effectiveness of the CatGrass model, a benchmark model based on the CatBoost algorithm was created, and the results were compared. Indeed, the comparison showed an improvement of 11% to 15% in R^2 values when using the CatGrass model. This increase in R^2 values highlights the enhanced predictive accuracy achieved through the novel feature engineering framework and GOA optimization. Additionally, significant reductions were observed in other performance metrics. These improvements collectively demonstrate the superiority of the CatGrass approach in providing more accurate and reliable predictions, particularly in the context of seismic response for low-rise RC frames.
- CatGrass also outperformed some popular ML algorithms (i.e., RF, SVM, and XGBoost) in terms of performance metrics; however, it fell behind as compared to the other ML algorithms based on computational time.
- Features such as *Sa*, *VSI* and *PGV* were found to be the strongest influences on the seismic response of both R3 and IR3 frames.

In this study, low-rise RC frames were considered and proposed CatGrass ML model is used for predicting of *MDR*. This situation can be considered a limitation of the study. However, future research could extend these findings by analyzing the effects of varying seismic scenarios and different structural configurations on the overall performance. A more in-depth exploration of different lateral strength ratios and periods would help refine the understanding of their influence on the nonlinear response. Furthermore, conducting comparative studies using alternative ML methods could complement the findings and broaden the scope of seismic risk assessment techniques.

Acknowledgements Not applicable.

Author contributions AD: Conceptualization, Writing – review & editing, Writing – original draft, Investigation, Visualization. SD: Conceptualization, Writing – review & editing, Writing – original draft, Investigation, Visualization. EKS: Conceptualization, Writing – review & editing, Writing – original draft, Investigation, Visualization.

Funding Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK). No funding was received for conducting this study.

Data availability Data will be made available on request.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Ancheta TD, Darragh RB, Stewart JP, Seyhan E, Silva WJ, Chiou BSJ et al (2014) NGAWest2 database. Earthq Spectra 30(3):989–1005. https://doi.org/10.1193/070913EQS197M
2. Ambraseys NN, Douglas J, Rinaldis D, Berge CT, Suhadolc P, Costa G, Sigbjornsson R, Smit P (2004) Dissemination of European strong-motion data, vol. 2. CD-ROM Collection. Engineering and Physical Sciences Research Council, Swindon.
3. Amezquita-Sanchez J, Valtierra-Rodriguez M, Adeli H (2020) Machine learning in structural engineering. Sci Iran, Civ Eng Trans 27(6):2645–2656. https://doi.org/10.24200/sci.2020.22091
4. Aljarah I, Al-Zoubi AM, Faris H et al (2018) Simultaneous feature selection and support vector machine optimization using the grasshopper optimization algorithm. Cogn Comput 10:478–495. https://doi.org/10.1007/s12559-017-9542-9
5. Araújo M, Macedo L, Marques M, Castro JM (2016) Code-based record selection methods for seismic performance assessment of buildings. Earthq Eng Struct Dyn 45(1):129–148. https://doi.org/10.1002/eqe.2620
6. ASCE 7–16 (2016) Minimum design loads for buildings and other structures. American Society of Civil Engineers, Reston.
7. Asgarkhani N, Kazemi F, Jakubczyk-Gałczynska A, Mohebi B, Jankowski R (2024) Seismic response and performance prediction of steel buckling-restrained braced frames using machine-learning methods. Eng Appl Artif Intell 128:107388. https://doi.org/10.1016/j.engappai.2023.107388
8. Asteris PG, Repapis CC, Repapi EV, Cavaleri L (2017) Fundamental period of infilled reinforced concrete frame structures. Struct Infrastruct Eng 13(7):929–941. https://doi.org/10.1080/15732479.2016.1227341
9. Ay ÖB, Erberik MA (2008) Vulnerability of Turkish low-rise and mid-rise reinforced concrete frame structures. J Earthq Eng 12(2):2–11. https://doi.org/10.1080/13632460802012687
10. Azizi M, Shishehgarkhaneh MB, Basiri M, Moehler RC, Fang Y, Chan M (2023) Wolf-Bird optimizer (WBO): a novel metaheuristic algorithm for building information modeling-based resource tradeoff. J Eng Res. https://doi.org/10.1016/j.jer.2023.11.024
11. Azizi M, Zhou A (2024) Optimized machine learning approach for structural response prediction using wolf-bird optimizer. Structures 65:106691. https://doi.org/10.1016/j.istruc.2024.106691
12. Box GE, Cox DR (1969) An analysis of transformations. J Roy Stat Soc: Ser B (Methodol) 26:211–243
13. Breiman L (1996) Bagging predictors. Mach Learn 24:123–140. https://doi.org/10.1007/BF00058655
14. Chandola V, Banerjee A, Kumar V (2009) Anomaly detection: a survey. ACM Comput Surv 41:1–58. https://doi.org/10.1145/1541880.1541882
15. Chen T, Guestrin C (2016) XGBoost: A scalable tree boosting system. 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining 785–794. https://doi.org/10.1145/2939672.2939785.
16. Cheng MY, Prayogo D (2014) Symbiotic organisms Search: a new metaheuristic optimization algorithm. Comput Struct 139:98–112. https://doi.org/10.1016/j.compstruc.2014.03.007
17. Cramer H (1946) Mathematical methods of statistics. Princeton Univ, Princeton, NJ
18. Demir A (2022) Investigation of the effect of real ground motion record number on seismic response of regular and vertically irregular RC frames. Structures 39:1074–1091. https://doi.org/10.1016/j.istruc.2022.03.091
19. Demir A, Sahin EK, Demir S (2024) Advanced tree-based machine learning methods for predicting the seismic response of regular and irregular RC frames. Structures 64:106524. https://doi.org/10.1016/j.istruc.2024.106524

20. Demir S, Sahin EK (2022) Comparison of tree-based machine learning algorithms for predicting liquefaction potential using canonical correlation forest, rotation forest, and random forest based on CPT data. Soil Dyn Earthq Eng 154:107130. https://doi.org/10.1016/j.soildyn.2021.107130
21. Demir S, Sahin EK (2023) Predicting occurrence of liquefaction-induced lateral spreading using gradient boosting algorithms integrated with particle swarm optimization: PSO-XGBoost, PSO-LightGBM, and PSO-CatBoost. Acta Geotech 18:3403–3419. https://doi.org/10.1007/s11440-022-01777-1
22. Demir A, Palanci M, Kayhan AH (2024) Evaluation the effect of amplitude scaling of real ground motions on seismic demands accounting different structural characteristics and soil classes. Bull Earthquake Eng 22:365–393. https://doi.org/10.1007/s10518-023-01780-1
23. Demir A, Kayhan AH, Palanci M (2023) Response- and probability-based evaluation of spectrally matched ground motion selection strategies for bi-directional dynamic analysis of low-to mid-rise RC buildings. Structures 58:105533. https://doi.org/10.1016/j.istruc.2023.105533
24. Demertzis K, Kostinakis K, Morfidis K, Iliadis L (2023) An interpretable machine learning method for the prediction of R/C buildings' seismic response. J Build Eng 63:105493. https://doi.org/10.1016/j.jobe.2022.105493
25. Dong J, Zeng W, Lei G, Wu L, Chen H, Wu J, Srivastava AK (2022) Simulation of dew point temperature in different time scales based on grasshopper algorithm optimized extreme gradient boosting. J Hydrol 606: 127452 https://doi.org/10.1016/j.jhydrol.2022.127452.
26. EUROCODE-8 (2004) Design provisions for earthquake resistance of structures, Part 1: General rules, seismic actions and rules for buildings. European Committee for Standardization, Brussels.
27. Fausett LW (1994) Fundamentals of Neural Networks: Architectures. Prentice-Hall, Algorithms and Applications
28. Feurer M, Hutter F (2019) Hyperparameter Optimization. In: Hutter F, Kotthoff L, Vanschoren J (eds) Automated Machine Learning. The Springer Series on Challenges in Machine Learning. Springer, Cham
29. Friedman JH (2001) Greedy function approximation: A gradient boosting machine. Ann Stat 29:1189–232. https://www.jstor.org/stable/2699986.
30. Goldberg DE (1989) Genetic Algorithms in Search. Optimization, and Machine Learning, 1st edn. Addison Wesley Pub, Boston, Massachusetts, USA
31. Geurts P, Ernst D, Wehenkel L (2006) Extremely randomized trees. Mach Learn 63:3–42. https://doi.org/10.1007/s10994-006-6226-1
32. Giordano N, De Luca F, Sextos A (2021) Analytical fragility curves for masonry school building portfolios in Nepal. Bull Earthq Eng 19:1121–1150. https://doi.org/10.1007/s10518-020-00989-8
33. Hamidia M, Kaboodkhani M, Bayesteh H (2024) Vision-oriented machine learning-assisted seismic energy dissipation estimation for damaged RC beam-column connections. Eng Struct 301:117345. https://doi.org/10.1016/j.engstruct.2023.117345
34. Harirchian E, Lahmer T, Kumari V, Jadhav K (2020) Application of support vector machine modeling for the rapid seismic hazard safety evaluation of existing buildings. Energies 13(13):3340. https://doi.org/10.3390/en13133340
35. Harifi S, Khalilian M, Mohammadzadeh J, Ebrahimnejad S (2020) Optimizing a neuro-fuzzy system based on nature-inspired emperor penguins colony optimization algorithm. IEEE Trans Fuzzy Syst 28(6):1110–1124. https://doi.org/10.1109/TFUZZ.2020.2984201
36. Hatzigeorgiou GD, Liolios AA (2010) Nonlinear behaviour of RC frames under repeated strong ground motions. Soil Dyn Earthq Eng 30(10):1010–1025. https://doi.org/10.1016/j.soildyn.2010.04.013
37. Hekimoğlu B, Ekinci S (2018) Grasshopper optimization algorithm for automatic voltage regulator system. In: 5th international conference on electrical and electronic engineering (5th ICEEE) 152–156. IEEE.
38. Huu-Tai T (2022) Machine learning for structural engineering: a state-of-the-art review. Structures 38:448–491. https://doi.org/10.1016/j.istruc.2022.02.003
39. Hwang SH, Mangalathu S, Shin J, Jeon JS (2021) Machine learning-based approaches for seismic demand and collapse of ductile reinforced concrete building frames. J Build Eng 34:101905. https://doi.org/10.1016/j.jobe.2020.101905
40. Ibrahim HT, Mazher WJ, Ucan ON et al (2019) A grasshopper optimizer approach for feature selection and optimizing SVM parameters utilizing real biomedical data sets. Neural Comput Appl 31:5965–5974. https://doi.org/10.1007/s00521-018-3414-4
41. Kamel SR, Yaghoubzadeh R (2021) Feature selection using grasshopper optimization algorithm in diagnosis of diabetes disease. Inform Med Unlocked 26:100707. https://doi.org/10.1016/j.imu.2021.100707
42. Kazemi F, Asgarkhani N, Jankowski R (2023) Machine learning-based seismic response and performance assessment of reinforced concrete buildings. Arch Civ Mech Eng 23:94. https://doi.org/10.1007/s43452-023-00631-9
43. Kennedy J, Eberhart R (1995) Particle swarm optimization. IEEE International Conference on Neural Networks, Piscataway, NJ.
44. Kiani J, Camp C, Pezeshk S (2019) On the application of machine learning techniques to derive seismic fragility curves. Comput Struct 218:108–122. https://doi.org/10.1016/j.compstruc.2019.03.004
45. Kostinakis K, Athanatopoulou A, Morfidis K (2015) Correlation between ground motion intensity measures and seismic damage of 3D R/C buildings. Eng Struct 82:151–167. https://doi.org/10.1016/j.engstruct.2014.10.035

46. Lazaridis PC, Kavvadias IE, Demertzis K, Iliadis L, Vasiliadis LK (2022) Structural damage prediction of a reinforced concrete frame under single and multiple seismic events using machine learning algorithms. Appl Sci 12(8):3845. https://doi.org/10.3390/app12083845
47. Lewis C (1982) International and Business Forecasting Methods Butterworths: London.
48. Li K, Xie X, Zhou B, Huang C, Lin W, Zhou Y, Wang C (2024) Thickness regression for backfill grouting of shield tunnels based on GPR data and CatBoost & BO-TPE: a full-scale model test study. Undergr Space 17:100–119. https://doi.org/10.1016/j.undsp.2023.10.003
49. Liaw A, Wiener M (2002) Classification and regression by randomForest. R News 2(3):18–22
50. Liu H, Setiono R (1995) Chi2: Feature selection and discretization of numeric attributes. In: Proceedings of the International Conference on Tools with Artificial Intelligence (7th IEEE) 388–391.
51. Mafarja M, Aljarah I, Faris H, Hammouri AI, Al-Zoubi A, Mirjalili S (2019) Binary grasshopper optimisation algorithm approaches for feature selection problems. Expert Syst Appl 117:267–286. https://doi.org/10.1016/j.eswa.2018.09.015
52. Mangalathu S, Hwang SH, Choi E, Jeon JS (2019) Rapid seismic damage evaluation of bridge portfolios using machine learning techniques. Eng Struct 201:109785. https://doi.org/10.1016/j.engstruct.2019.109785
53. Mander JB (1983) Seismic design of bridge piers. research report 84–2, Department of Civil Engineering, University of Canterbury, Christchurch.
54. Mayer M (2023) outForest: Multivariate Outlier Detection and Replacement, R package version 1.0.1. https://CRAN.Rproject.org/package=outForest/. [accessed 13 March 2024].
55. Meral E (2024) Relationships between ground motion parameters and energy demands for regular low-rise RC frame buildings. Bull Earthquake Eng 22:2829–2865. https://doi.org/10.1007/s10518-024-01885-1
56. Mirrashid M, Naderpour H (2022) Computational intelligence-based models for estimating the fundamental period of infilled reinforced concrete frames. J Build Eng 46:103456. https://doi.org/10.1016/j.jobe.2021.103456
57. Mirjalili S, Lewis A (2016) The whale optimization algorithm. Adv Eng Softw 95:51–67. https://doi.org/10.1016/j.advengsoft.2016.01.008
58. Mirjalili S, Jangir P, Mirjalili SZ, Saremi S, Trivedi IN (2017) Optimization of problems with multiple objectives using the multi-verse optimization algorithm. Knowl Based Syst 134:50–71. https://doi.org/10.1016/j.knosys.2017.07.018
59. Morfidis K, Kostinakis K (2023) Seismic parameters' combinations for the optimum prediction of the damage state of R/C buildings using neural networks. Adv Eng Softw 106:1–16. https://doi.org/10.1016/j.advengsoft.2017.01.001
60. Noureldin M, Ali T, Kim J (2023) Machine Learning-based seismic assessment of framed structures with soil-structure interaction. Front Struct Civ Eng 17:205–223. https://doi.org/10.1007/s11709-022-0909-y
61. Palanci M, Senel SM (2019) Correlation of earthquake intensity measures and spectral displacement demands in building type structures. Soil Dyn Earthq Eng 121:306–326. https://doi.org/10.1016/j.soildyn.2019.03.023
62. Palanci M, Demir A, Kayhan AH (2023) Quantifying the effect of amplitude scaling of real ground motions based on structural responses of vertically irregular and regular RC frames. Struct 51:105–123. https://doi.org/10.1016/j.istruc.2023.03.040
63. Paudel S, Pudasaini A, Shrestha RK, Kharel E (2023) Compressive strength of concrete material using machine learning techniques. Clean Eng Technol 15:100661. https://doi.org/10.1016/J.CLET.2023.100661
64. Priestley MJN, Calvi GM, Kowalsky MJ (2007) Displacement-Based Seismic Design of Structures. IUSS Press, Pavia, Italy
65. Prokhorenkova L, Gusev G, Vorobev A, Dorogush AV, Gulin A (2018) CatBoost: unbiased boosting with categorical features. In: Proceedings of the 32nd International Conference on Neural Information Processing Systems, 6639–49.
66. Rafiei MH, Adeli H (2017) A novel machine learning-based algorithm to detect damage in high-rise building structures. Struct Des Tall Build 26(18):e1400. https://doi.org/10.1002/tal.1400
67. Rahman T, Hasan MH, Momin MF, Zheng P (2024) Data-driven approach to predict the fundamental period of steel-braced RC frames using stacked generalization machine learning models. Asian J Civ Eng 25(3):2379–2397. https://doi.org/10.1007/s42107-023-00914-9
68. Riza LS, Nugroho EP, Prabowo MBA, Junaeti E, Abdullah AG (2019) Metaheuristicopt: Metaheuristic for optimization. R package version 2.0. 0. https://doi.org/10.32614/CRAN.package.metaheuristicOpt.
69. Rodrigues H, Furtado A, Vila-Pouca N, Varum H, Barbosa AR (2018) Seismic assessment of a school building in Nepal and analysis of retrofitting solutions. Int J Civ Eng 16:1573–1589. https://doi.org/10.1007/s40999-018-0297-9
70. Russell SJ, Norvig P (2009) Artificial Intelligence: A Modern Approach. 3rd, edition. Pearson, Upper Saddle River
71. Sahin EK, Demir S (2023) Greedy-AutoML: a novel greedy-based stacking ensemble learning framework for assessing soil liquefaction potential. Eng Appl Artif Intell 119:105732. https://doi.org/10.1016/j.engappai.2022.105732
72. SAP2000 (2009) Integrated solution for structural analysis and design. Computers and Structures, Berkeley.
73. Saremi S, Mirjalili S, Lewis A (2017) Grasshopper optimisation algorithm: Theory and application. Adv Eng Softw 105:30–47. https://doi.org/10.1016/j.advengsoft.2017.01.004
74. Sreehari E, Babu LD (2024) Critical factor analysis for prediction of diabetes mellitus using an inclusive feature selection strategy. Appl Artif Intell 38(1):2331919. https://doi.org/10.1080/08839514.2024.2331919
75. Storn R, Price K (1997) Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. J Global Optim 11:341–359. https://doi.org/10.1023/A:1008202821328

76. Sun H, Burton HV, Huang H (2021) Machine learning applications for building structural design and performance assessment: state-of-the-art review. J Build Eng 33:101816. https://doi.org/10.1016/j.jobe.2020.101816
77. Sun S, Cao Z, Zhu H, Zhao J (2019) A survey of optimization methods from a machine learning perspective. IEEE Trans Cybern 50(8):3668–3681. https://doi.org/10.48550/arXiv.1906.06821
78. TBEC (2018). Turkish building earthquake code. Disaster and Emergency Management Presidency, Ankara.
79. Wang SC (2003) Artificial Neural Network. In: Interdisciplinary Computing in Java Programming. The Springer International Series in Engineering and Computer Science. 743. Springer, Boston, 81–100 https://doi.org/10.1007/978-1-4615-0377-4_5.
80. Yahiaoui A, Dorbani S, Yahiaoui L (2023) Machine learning techniques to predict the fundamental period of infilled reinforced concrete frame buildings. Structures 54:918–927. https://doi.org/10.1016/j.istruc.2023.05.052
81. Yang XS, He X (2013) Firefly algorithm: recent advances and applications. Int J Swarm Intell 1(1):36–50. https://doi.org/10.1504/IJSI.2013.055801
82. Yang L, Shami A (2020) On hyperparameter optimization of machine learning algorithms: theory and practice. Neurocomputing 415:295–316. https://doi.org/10.1016/j.neucom.2020.07.061
83. Yu Y, Li W, Li J, Nguyen TN (2018) A novel optimised self-learning method for compressive strength prediction of high performance concrete. Constr Build Mater 184:229–247. https://doi.org/10.1016/j.conbuildmat.2018.06.219

Authors and Affiliations

Ahmet Demir[1] · Selçuk Demir[1] · Emrehan Kutlug Sahin[1]

✉ Ahmet Demir
ahmetdemir@ibu.edu.tr

Selçuk Demir
selcukdemir@ibu.edu.tr

Emrehan Kutlug Sahin
emrehansahin@ibu.edu.tr

[1] Department of Civil Engineering, Bolu Abant Izzet Baysal University, 14030 Bolu, Türkiye